

1461. Check If a String Contains All Binary Codes of Size K

Solved 

Medium

 Topics

 Companies

 Hint

Given a binary string `s` and an integer `k`, return `true` if every binary code of length `k` is a substring of `s`. Otherwise, return `false`.

Example 1:

Input: `s = "00110110"`, `k = 2`

Output: `true`

Explanation: The binary codes of length 2 are "00", "01", "10" and "11". They can be all found as substrings at indices 0, 1, 3 and 2 respectively.

Example 2:

Input: `s = "0110"`, `k = 1`

Output: `true`

Explanation: The binary codes of length 1 are "0" and "1", it is clear that both exist as a substring.

Example 3:

Input: `s = "0110", k = 2`

Output: `false`

Explanation: The binary code "00" is of length 2 and does not exist in the array.

Constraints:

- `1 <= s.length <= 5 * 105`
- `s[i]` is either `'0'` or `'1'`.
- `1 <= k <= 20`

Python:

class Solution:

def hasAllCodes(self, s: str, k: int) -> bool:

req = 1 << k

seen = [False] * req

mask = req - 1

h = 0

for i, ch in enumerate(s):

h = ((h << 1) & mask) | (ord(ch) & 1)

if i >= k - 1 and not seen[h]:

seen[h] = True

req -= 1

if req == 0:

return True

return False

JavaScript:

var hasAllCodes = function(s, k) {

let total = 1 << k;

let seen = new Array(total).fill(false);

let mask = 0;

let limit = total - 1;

```

for (let i = 0; i < s.length; i++) {
    mask = ((mask << 1) & limit) | (s[i] === '1');
    if (i >= k - 1 && !seen[mask]) {
        seen[mask] = true;
        total--;
        if (total === 0) return true;
    }
}
return false;
};

```

Java:

```

class Solution {
    public boolean hasAllCodes(String s, int k) {
        int req = 1 << k;
        boolean[] seen = new boolean[req];
        int mask = req - 1;
        int hash = 0;

        for (int i = 0; i < s.length(); ++i) {
            hash = ((hash << 1) & mask) | (s.charAt(i) & 1);

            if (i >= k - 1 && !seen[hash]) {
                seen[hash] = true;
                req--;
                if (req == 0) return true;
            }
        }

        return false;
    }
}

```